# Linux Cheat Sheet

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## Linux Terminal

### General Commands

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|---|---|
| `clear` | Clear the terminal screen. |
| `history` | Display the command history. |
| `cal` | Display a calendar. |
| `uptime` | Display how long the system has been running. |
| `man` | Display the manual for a command. (Tap q to quit the manual) |
| `date` | Display the current date and time. |
| `date "+Y-m-d 00:00:00"` | set the date and time. |

### File Management Commands

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| `cd` | Change directory. |
| `cd ..` | Move to the previous directory. |
| `cd ~` | Move to the home directory. |
| `cd /` | Move to the root directory. |
| `ls` | List files in the current directory. |
| `ls -l` | List files with detailed information. |
| `ls -a` | List all files, including hidden ones. |
| `pwd` | Print the current working directory. |
| `open` | Open a file with the default application. |
| `mkdir` | Create a new directory. |
| `rm` | Remove a file. |
| `rm -r` | Remove a directory and its contents. |
| `cp` | Copy a file or directory. |
| `mv` | Move or rename a file or directory. |
| `touch` | Create a new empty file. |
| `cat` | Display the contents of a file. |
| `tail` | Display the last few lines of a file. |
| `tail -f` | Continuously display new lines added to a file. |
| `head` | Display the first few lines of a file. |
| `stat` | Display detailed information about a file. |

### OS Commands

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|---|---|
| `sudo` | Execute a command with superuser privileges. |
| `sudo -l` | List permitted commands. |
| `whoami` | Display the current user. |
| `who` | Display all logged-in users. |
| `id` | Display user ID and group ID information. |
| `id -u` | Display the user ID. |
| `id -g` | Display the group ID. |
| `ps` | Display currently running processes. |
| `kill` | Terminate a process by its PID. |
| `kill -9` | Forcefully terminate a process by its PID. |
| `reboot` | Restart the system. |
| `poweroff` | Shut down the system. |
| `useradd` | Add a new user. |
| `userdel` | Delete a user. |
| `passwd` | Change a user's password. |
| `su - user` | Switch to another user. |
| `chown user file` | Change the owner of a file to `user`. |
| `chown user:group file` | Change the owner and group of a file. |
| `chmod flag file -u tom` | Change the permissions of `file` for user `tom`. |
| `chmod +x file` | Add execute permission to a file. |
| `chmod +r file` | Add read permission to a file. |
| `chmod +w file` | Add write permission to a file. |
| `chmod -x file` | Remove execute permission from a file. |
| `chmod -r file` | Remove read permission from a file. |
| `chmod -w file` | Remove write permission from a file. |

### Systemd Services

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| `systemctl start service` | Start a service. |
| `systemctl stop service` | Stop a service. |
| `systemctl restart service` | Restart a service. |
| `systemctl status service` | Check the status of a service. |
| `systemctl enable service` | Enable a service to start on boot. |
| `systemctl disable service` | Disable a service from starting on boot. |
| `systemctl daemon-reload` | Reload the systemd manager configuration |
| `systemctl reset-failed service` | Reset the failed state of a service. |
| `journalctl -u service` | View logs for a specific service. |
| `journalctl -n 100 -u service` | View the last 100 log entries for a specific service. |
| `--no-pager` | Disable paging of log output. |

## Shell Scripting

### operators

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| > | Redirect output to a file, overwriting the file if it exists. |
| >> | Redirect output to a file, appending to the file if it exists. |
| < | Redirect input from a file. |
| \| | Pipe the output of one command to another command. |
| `&&` | Execute the next command only if the previous command succeeded. |

### Basic Syntax

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|---|---|
| `#!/bin/bash` | Shebang line to specify the script interpreter. |
| `echo` | Print text to the terminal. |
| `read` | Read input from the user. |
| `if` | Start of a conditional statement. |
| `elif` | Else if condition in a conditional statement. |
| `else` | Else condition in a conditional statement. |
| `fi` | End of a conditional statement. |
| `for` | Start of a for loop. |
| `while` | Start of a while loop. |
| `do` | Start of the loop body. |
| `done` | End of a loop. |
| `$0` | The name of the script. |
| `$1, $2, ...` | Positional parameters for script arguments. |
| `grep` | Search for a pattern in a file or input. |
| `exec` | Execute a command. |
| `exit` | Exit the script. |

### Networking Commands

The below commands requires the `net-tools` package to be installed on the system.

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|---|---|
| `ifconfig` | Displays IP address. |
| `ping` | Returns packet loss between two the device and a destination. |
| `traceroute` | Traces the path packets take to a destination. |
| `nslookup` | Returns IP address information for a domain name. |
| `arp -a` | Displays the ARP table, which maps IP addresses to MAC addresses. |
| `arp myIPa` | Displays the ARP table for the specific IP address. |
| `curl` | Transfers data from or to a server, using various protocols. |
| `curl ifconfig.me` | Display machine's public IPv4 address. |

## SSH connection

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| `ssh -i ~/.ssh/myKey user@ipv4` | Connect to a remote host via SSH. |
| `exit` | Exit the SSH session. |
| `Ctrl + D` | Exit the SSH session. |

# package managers

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|---|---|
| `apt` | Debian-based(include ubuntu) package manager. |
| `apk` | Alpine Linux package manager. |
| `yum` | Red Hat-based package manager. |
| `dnf` | Fedora-based package manager. |
| `pacman` | Arch Linux package manager. |
| `brew` | Alternative package manager for macOS and many Linux distributions. |

## apk

| | |
|---|---|
| `update` | Update repository indexes. |
| `upgrade` | Install upgrades available from repositories . |
| `list` | List all available packages. |
| `add pkg` | Install a package named `pkg`. |
| `del pkg` | Remove a package named `pkg`. |
| `search pkg` | Search for a package named `pkg`. |
| `info pkg` | Display detailed information about a package named `pkg`. |

## apt

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|---|---|
| `update` | Update the package list. |
| `upgrade` | Upgrade all installed packages to their latest versions. |
| `list` | List all available packages. |
| `list --upgradable` | List all packages that can be upgraded. |
| `install pkg` | Install a package named `pkg`. |
| `remove pkg` | Remove a package named `pkg`. |
| `search pkg` | Search for a package named `pkg`. |